Sales Performance Analysis: Leveraging Data Insights for Strategic Business Decision-Making

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Data Visualisation (CMP020L013) Assessment Report

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1. Introduction

The data-driven decision making has become a necessity in an increasingly competitive retail market to maintain and enhance profits, customer loyalty and build business growth techniques. Price promotions and discounting are the core for focusing on consumer purchases, but their effectiveness depends on various factors like timing, type of product and customers.

Recent research by [1] introduced a customer-centric market basket analysis approach, which integrates customer segmentation with association rules to reveal more co-purchasing items which allowed more personalized bundling leading to higher returns. Complementing this approach, [2] proposed a three-phase framework comprising of "Data Acquisition, Analytics and Insight Generation and Strategic Implementation" for small enterprises mainly focusing on predictive analysis and real-time decision systems to improve operations and scale their decision intelligence. [3] further validated that companies that leverage data science techniques such as predictive modelling, segmentation analysis and data visualization have significant positive effect in revenue, cost efficiency and customer satisfaction.

Building on the frameworks, this report examines a dataset of 9993 retail transactions containing of diverse products and customer segments and geographic regions. The analysis is structed around three critical business objectives:

- Revenue Predictability: identifying how temporal sales, profit patterns and seasonality can affect the business performance.
- Profit Maximization: Understanding the threshold at which the discounts transitions from profit to loss-making.
- Customer Value Enhancement: Finding product combinations with the help of market basket analysis to inform cross-selling recommendations.

Visual tools such as scatterplots, boxplots and segmented association rules are implemented to uncover the insights. This report combines recent research with real-world sales data to offer practical guidance for retail decision-makers. The goal is to move away from using discounts in a one-size-fits-all way. Instead, it promotes a methodology that uses data to predict customer behaviour, target the right buyers and enhance profits. This approach helps boost short-term sales while also looking out for long-term business growth and brand value.

2. Data Preparation and Exploratory Data Analysis (EDA)

The quality of data is the core of any trustworthy business and so adapting effective and comprehensive data preparation and critical exploratory data analysis of the clean and accurate data is important for generating useful insights for better business decisions. This section of the report will establish a deep understanding of the dataset; it's quality and structure with the help of descriptive statistics and distribution plots.

The dataset had no missing values in all the 19 attributes and comprised of only one duplicate entry which was then removed from the dataset. To implement time-series analysis and to ensure accurate temporal sequencing, the date was formatted and standardized to datetime format.

The descriptive statistical analysis was conducted and revealed notable patterns. Order and ship dates span from January 2014 to early January 2018, and half of the orders happened after June 2016. Sales had a median of $54.48 and a mean of $229.85, showing that big orders had an impact on overall revenue. Profit varied a lot, some orders lost money while others earned a lot. This suggests that there are opportunities to keeping costs in check and what works best for the growth of the business. Quantity was quite consistent with a median of 3 to 4 units and maximum reaching to 14 and Discounts level were also varied with half of them under 20% or less and a peak of 80%.

2.1 Outlier Detection

Outlier detection detects points or values in the dataset which are extremely far from the distribution. In business scenarios, these kinds of values may depict rare but significant situations like bulk orders or special offers and discounts.

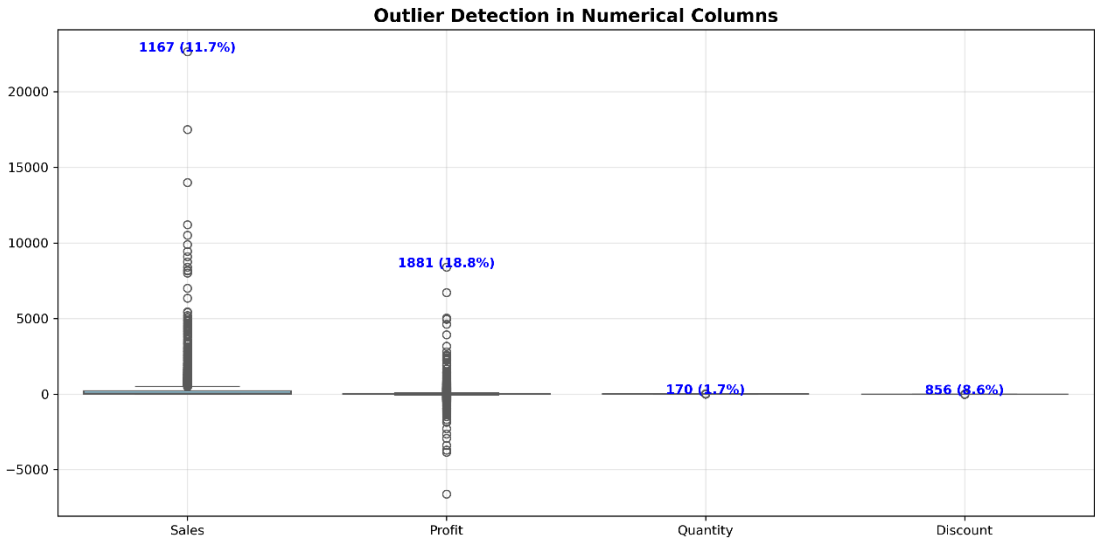

Figure 1: Outlier detection boxplots

The outlier detection analysis was conducted using boxplots across the numerical columns and in figure 1 reveals the distribution of outliers among the numerical columns in which Sales showed 1167 (11.7%), Profit showed 1881 (18.8%) extreme values, quantity containing 170 (1.7%) outliers and Discount displaying 856 (8.6%) outliers. This revelation arose several significant insights from business perspective that is better to be kept rather than removal as they likely represent extremely high value transactions or huge orders and crucial for better analysis.

2.2 Distribution Analysis

Understanding the spread of the numerical variables provide reflections into customer and product patterns. This section uses histograms to find out if sales, profit margins, quantity and discount rates have a normal or skewed distribution.

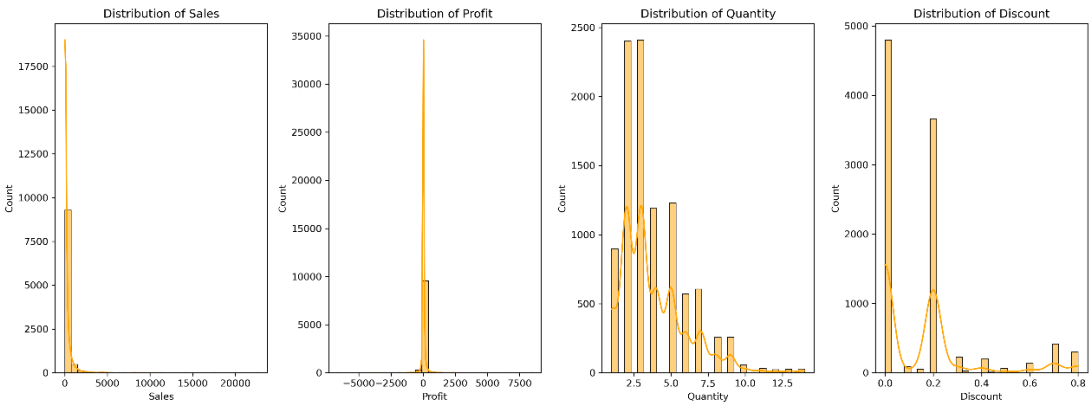

Figure 2: Distribution of Sales, Profit, Quantity and Discount

The histograms for **sales** and **profit** are positively skewed, with majority of transactions are concentrated below $2500 and some high value sales, forming a long tail of large orders. In contrast, **Quantity** shows a near normal distribution, indicating a consistent ordering pattern. The **Discount** variable's distribution is bimodal with cluster around 0% and 20%, with maximum discount reaching up to 80% which shows varied promotional strategies.

2.3 Category and Regional Analysis

This kind of analysis can depict what are the businesses strong and weak points. This section analyses categorical trends to know which product and regions contribute the highest to quantity and profitability.

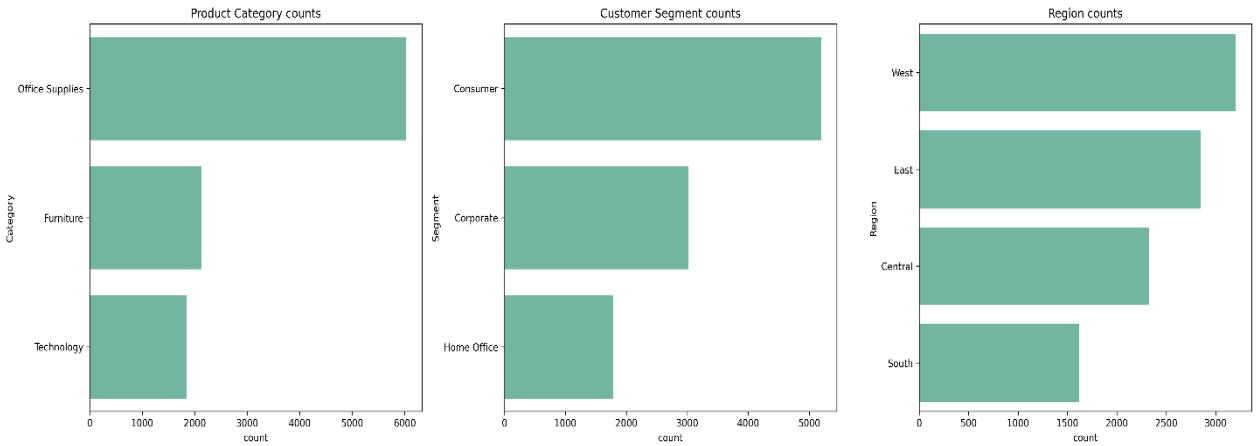

Figure 3: transaction count by region and customer segment

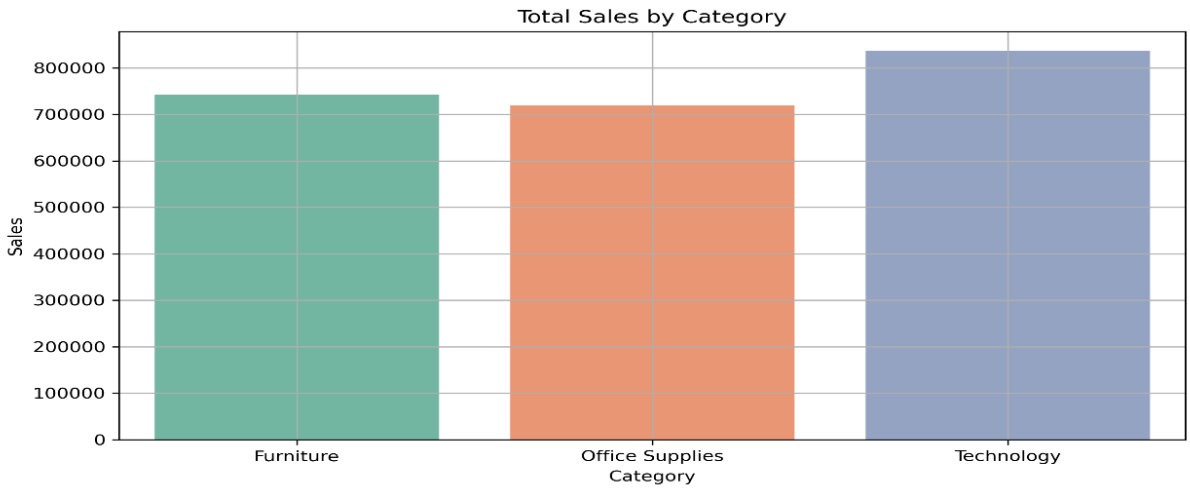

Figure 4: Profit contribution by product category

The figure 3 and 4 reveal key findings like:

- Technology leads in overall profit but has lower transaction volume.
- Office Supplies have the highest number of count but yields less profit and furniture has comparatively high sales with different margins.
- In the regional and segment distribution, the **West** region is the strongest market with more than around 32% of transactions.
- **Consumer segment** dominates, followed by corporate clients and home office.

2.4 Correlation Matrix

Correlation matrix helps in identifying linear relationships between numerical variables.

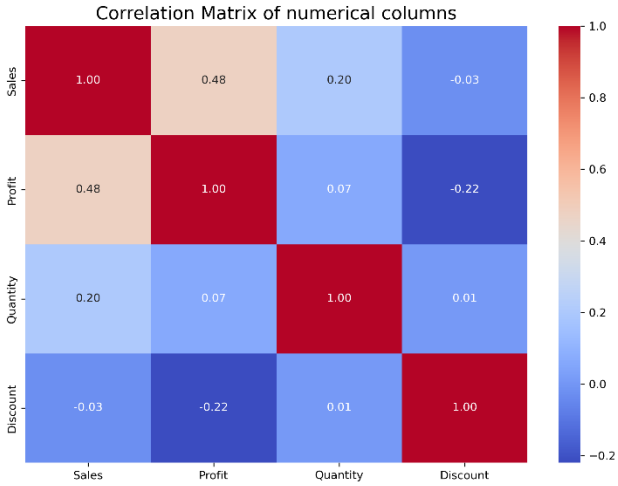

Figure 5: Correlation matrix

The key finding here in figure 5 is the negative correlation of **-0.22** between *Discount* and *Profit* which suggest that higher discounts lead to low profits and sets the foundation for more detailed examination of this relationship. The correlation serves as a foundation for the problem explored in section 3.2 of the report, guiding more detailed examination of discount sensitivity and profitability in different categories.

3. Research Questions and Integration of Domain Knowledge

The foundation of this analysis is built around three research questions that focuses on important aspects a business, namely seasonality, discount-profitability relationship and customer purchasing behaviour. Every sub-section explores one of these aspects integrate the domain knowledge, it's relevance and uses data visualization to find insights.

3.1 What quarterly and monthly pattern in Sales and Profit can inform Inventory Optimization and Marketing campaign timing?

Justification and Domain Context:

Seasonality has been recognised as a core factor in retail environment since a long time, especially for inventory planning, resource allocation and campaign timing. Analysing sales and profit over time would help businesses find a pattern and plan business operations accordingly. These patterns could be based on many factors including weather, holidays and festivals [4]. The dataset's inclusion of columns like order date, sales and profit enabled the feasibility of conducting the analysis.

The hypothesis for conducting this analysis was that sales and profit increased during winters, especially in November and December, which is also festival time.

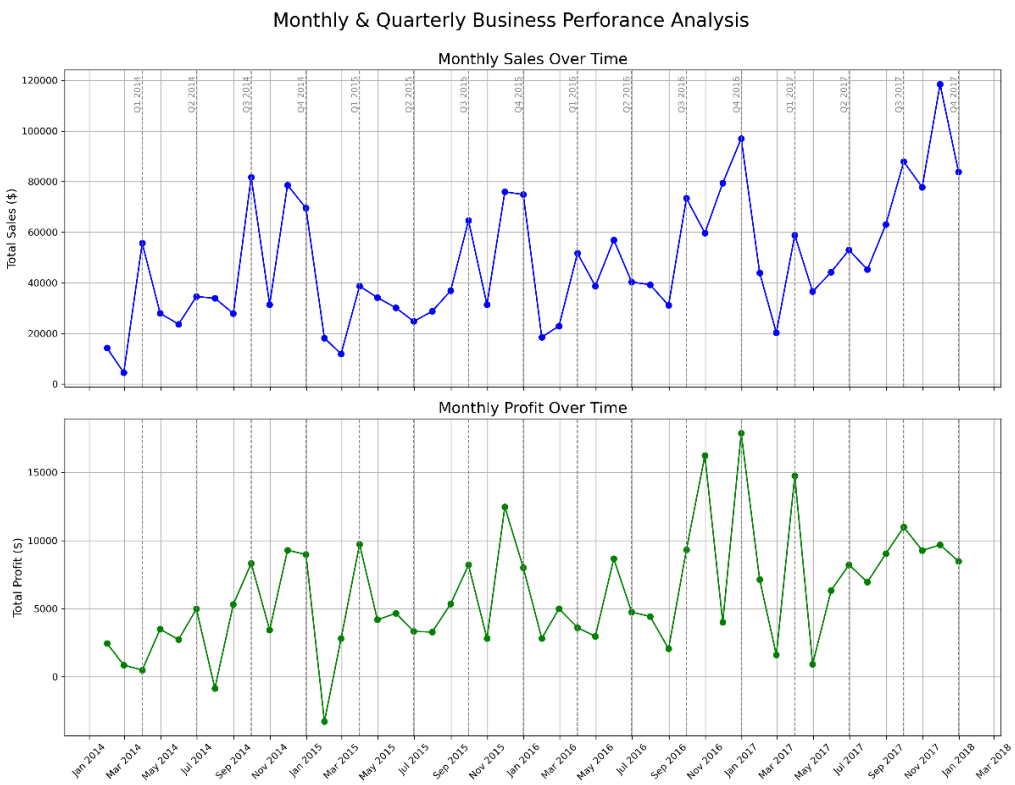

Figure 6: Time-series visualization showing monthly and quarterly sales and profit analysis

Observation:

The figure 6 shows monthly sales and profit from **January 2014** to **February 2018**, along with quarterly markers emphasizing the seasonal patterns. Sales, as visible, depicts an **upward trend** with high peaks in **November – December** every year, which aligns with my hypothesis and has festivals like **Thanksgiving** and **Christmas**. The highest occurs in December 2017, when profits surpass $100,000.

Profit shows a more fluctuating and complex trend. Early 2014 and mid 2015 depicts negative profit which is possibly due to heavy discounts. On the other hand, the time from **July 2016** to **March 2017** has **high profit-making** months, exceeding $15,000.

This temporal analysis suggests that the business operates in consistent and predictable environment, but high sales does not always comprehend to high profits. This makes a pathway to focus on effective cost management, discount strategies and product margins. Moreover, the fluctuations between sales peak and profit dips rises the importance of understanding the cross-functional coordination particularly between marketing, finance and operations team to ensure that it results into sustainable financial outcomes. Above all, the business shows a positive trend, with **Q4** being the highest revenue generative quarter.

3.2 What is the relationship between Discount and Profitability, and is there a threshold point where higher discounts lead to losses?

Justification and Domain Context:

According to [5], promotion and price discounts is a way of attracting people towards their product and increase sales. It is also a good and effective way of dealing with competitors. The researchers also suggest that even though they are an excellent way of marketing communication to boost purchases, excessive discounting may affect profitability, weaken the brand equity and may hinder growth [5], which matches with the hypothesis that profitability declines when discounts exceed **20%**. Additionally, [1] further supports that while discounts can be very effective in increasing purchases, their impact on profit in highly '**context-dependent'**.

Observation:

A scatterplot of transactions in figure 7 confirms the reduction in profit with increasing discounts, especially the **technology** category as even a moderate discount levels resulted in margin erosion. The visual density of negative profit transactions increases rapidly after the 20-25% discount mark, supporting the presence of threshold effect.

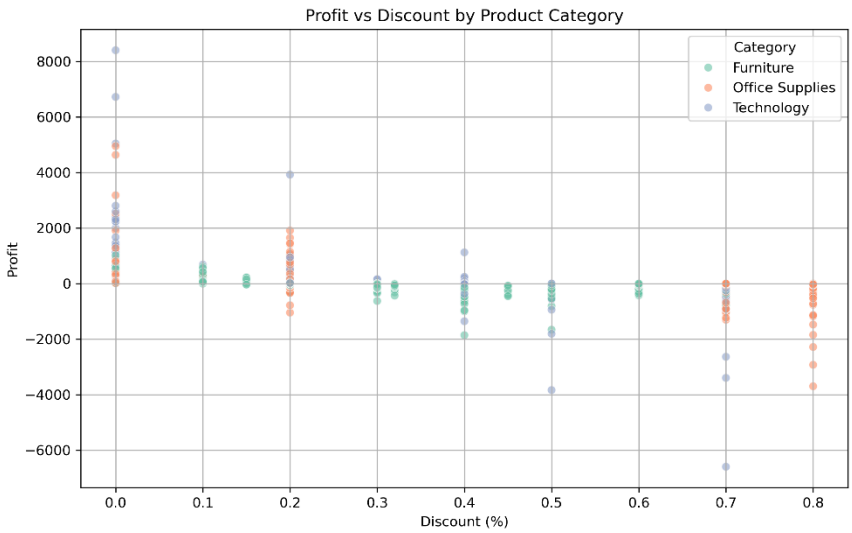

Figure 7: Scatter plot depicting profit vs discount by product category

A more granular analysis depicted using boxplot in the below figure 8, supports this conclusion as well.

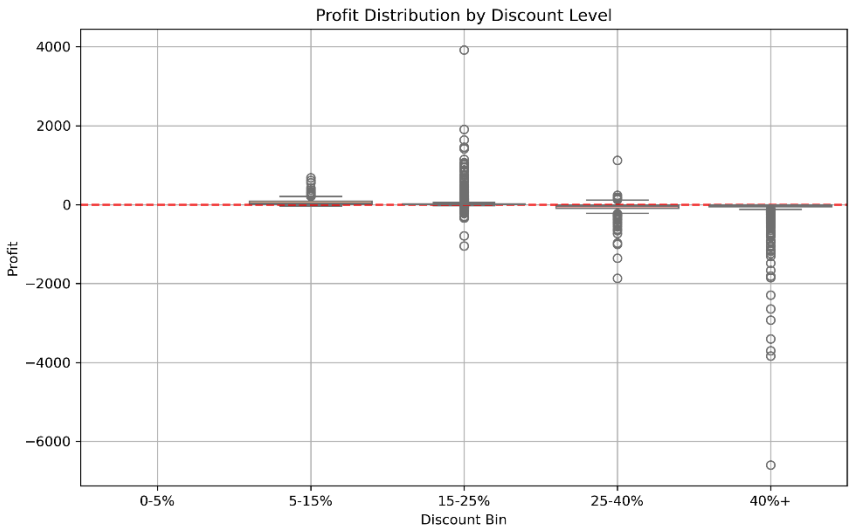

Figure 8: Profit distribution by discounts level

The median profit remains positive for transactions with 0 – 20% discount level and turns negative in the discounts above 30% and finally get highly skewed for discounts above 40%. These findings emphasize the importance of category-specific pricing strategies. Technology, because of tight margins, cannot have discounts more than 10-15% and Office-supplies shows more resilience for up to 25%. Therefore, the visuals suggest that indiscriminate discounting risks profitability and should be handled with more data-driven strategies.

To complement the scatterplot and the boxplot, a stacked bar chart was developed to visualize the distribution of profitable versus the loss-making orders in different discount ranges.

Figure 9: Stacked bar chart showing profitability distribution by discount levels

The bar chart in figure 9 shows that at 0-15% even though the number of orders is modestly less, there are comparatively more profit-making orders. Interestingly, the 15-25% range has a high volume of orders reaching around 3200, most of which are profitable. This confirms that, thoughtful discounting within this band can stimulate sales to a great extent.

In contrast, after 25% discount, the loss-making orders (the red bars) starts dominating and the volume of orders also starts reducing. Thus, the bar chart, due to it's color-coding and volume-based segmentation, reinforces the findings from the scatterplot and boxplot.

Taken together, these visualizations consistently confirm the threshold hypothesis which is discounting beyond ~20% leads to diminishing returns and losses. In order to prevent this loss, the company must avoid one-size-fits-all discounts and tailor discounts based on product margins to balance volume stimulation with maintaining the profitability.

3.3 What product combinations are most frequently purchased together and how can that help in managing promotions and offer?

Justification and Domain Context:

The analysis of asking the fundamental question of which products are frequently bought together is known as Market Basket Analysis and provides insights for cross-selling strategies, inventory optimization and most importantly customer behaviour understanding [1]. The analysis is also valuable for retail operations in order to optimize product management and promotional bundling [5]. As highlighted by [1], market basket analysis improves the relevance and strength of identified product combinations, allowing for more and better personalized shopping experience.

The integration of domain knowledge in market basket analysis (MBA) comes from the already established retail merchandising ethics, where understanding product complementarity can drive increased transactions and customers. By applying this approach in this report, the aim is to reveal high-impact item pairs that can be used to make smarter and evidence based offers for the consumers.

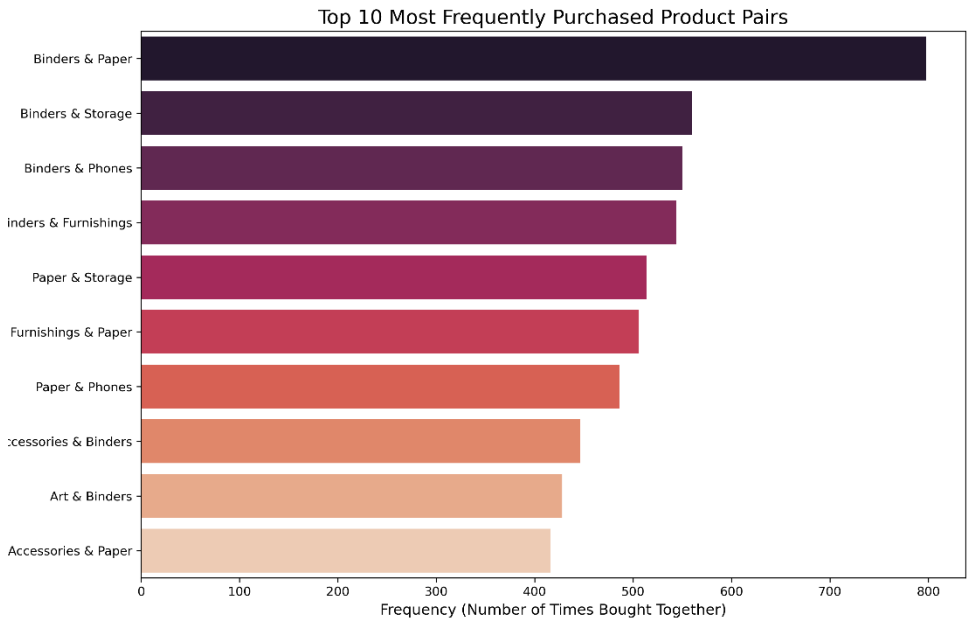

Figure 10: Bar chart showing top 10 most frequently bought together pairs

In the figure 10, Binders and Paper emerged as the most frequently bought product combination, appearing together more than 800 transactions. The top 10 product pairs reveal a strong office supply ecosystem where customers consistently purchase complementary items:

- Binders pair with storage, phones and furnishings
- Paper products connect across multiple categories including storage, phones and art supplies.

The network analysis visualization provided deeper insights into product relationship structures. It showed that binders serve as a central hub in the co-purchasing network, connecting to other multiple products, as shown in figure 11. Also, paper products show high combinations in the network with various office supplies.

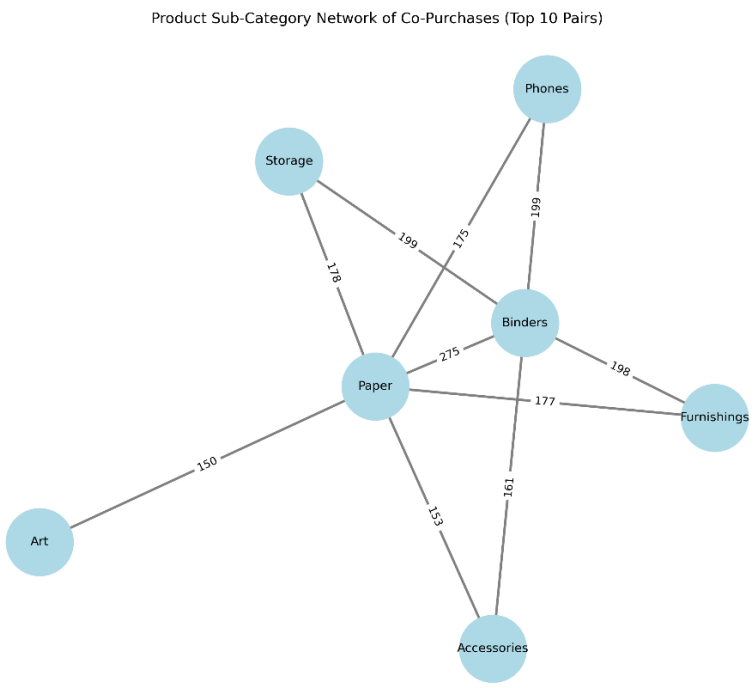

Figure 11: Network of items highly co-purchased together.

These identified product relationships gives a roadmap for targeted cross-selling initiatives. The sales team can use these patterns to recommend such partnerships at the point of sale, potentially **increasing revenue**. The analysis supports bundling strategies that align with natural customer purchasing behaviours. The display or **merchandising** of products either physical or digital can be

optimized based on these occurrences. A few recommendations such as developing marketing events that uses the role of binders and paper products to encourage a larger audience, monitoring a real-time analysis of tracking the co-purchases and optimizing product placements can highly provide better customer shopping experiences and in turn increase sales.

The limitation of market basket analysis is that it catches association, not causation, so it is difficult to say which item was selected first and whether it was intentional. Nonetheless, it demonstrates that customer purchasing behaviour follows a typical predictable pattern, particularly within office supplies, indicating opportunities for strategic business optimization and enhanced customer value delivery.

4. Findings and Conclusions

The detailed visual and statistical analysis of the sales dataset has unleashed several key insights, each in line with the core business operations including demand planning, price optimization, customer relationship management and promotions. After combining temporal, financial and behavioural perspectives, the findings provide not only descriptive observations but also solutions that can enhance both short-term and long-term operational decisions.

- Seasonal analysis demonstrated **Q4** as the **high-revenue** window.
- Discount versus profitability analysis showed how **beyond 20-25%** discount levels, the profit starts **declining**, especially for Technology category.
- Market basket analysis uncovered important bundling suggestions, majorly centring around **Binders**.

A major finding relates to **temporal sales and profit trends**, which highly supports the hypothesis of seasonal growth of revenue in Q4. The time-series analysis (figure 6) clearly shows that November and December regularly outperform all the other months in terms of quantity and sales. On the other hand, understanding the cause of **declining trend in Q1** which is a post-holiday time can assist in managing overstock risks and adjusting the sales targets.

Another major task is understanding the relationship between discount and profitability, which definitely challenges the old assumption that high discounts lead to higher performance. Across multiple plots including scatterplot (figure 7), boxplot (figure 8) and a stacked bar chart (figure 9), there is clear evidence that any discount above the threshold of **25%** leads to loss-making orders. Therefore, adopting category-specific offers, strategic discount timings, majorly during Q4 due to holidays, providing free shipping offers at times could enhance the business performance while also maintaining the competition in the market.

With the help of frequent item pair chart and the network graph (figures 10 and 11), the MBA reveals that Office Supplies, especially Binders, is the product which is co-purchased in almost more than 800 transactions. Further recommendations like providing **Office Starter kit**, or **in-store product placements** and formations like **'frequently bought together'**, like on other e-commerce platforms such as Amazon, could highly enhance the company's overall performance. This kind of visual analysis also provides the company with strong capacity to implement data-driven promotional events to improve customer satisfaction.

All the three research questions form a comprehensive framework which integrates operational, pricing and marketing decision based on data-driven methods. For the real-world business functions, there is a high impact of seasonality on pricing strategy and some bundling particularly targeting and improving the discounting optimization in certain quarters or time frame. From a methodological perspective, the visualizations were carefully selected to align with the nature of the question such as line plot for trends, scatterplot and boxplots for numerical variables and bar chart and networks for categorical relationships. Each visual is effectively interpreted in context and tied to the relevant business decision-making.

However, the analysis also encountered some limitations. The dataset represents historic and a specific time period data which may not capture trends beyond that four-year window. The analysis does not include the external economic factors, competitive actions or various PR events that may influence the revealed patterns. In future, including real-time analysis and machine learning based forecasting would allow for deeper strategic insights.

In conclusion, this analysis shows the power of data visualization as a strategic decision-making method. It provides impactful evidence that businesses which uses data literacy can outperform competitors by understanding basic operations like when to promote, how to price and not based on mere assumptions, but on patterns from the data itself. The recommendations coming from this analysis ranges from seasonal inventory planning to discount optimization and bundling design. With appropriate and thoughtful investments in infrastructure and operational activities, these findings can turn into improvements in profitability and customer satisfaction.

References

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Appendices

Appendix A: Statistical Summary Tables

This is the detailed statistical summary of the variables including measures of central tendency, dispersion and distribution characteristics.

Table 1: Descriptive statistics – Dates

Statistic	Order Date	Ship Date
Count	9,993	9,993
Mean	2016-04-30 01:53:33	2016-05-04 00:53:19
Min	2014-01-03 00:00:00	2014-01-07 00:00:00
25%	2015-05-23 00:00:00	2015-05-27 00:00:00
50%	2016-06-26 00:00:00	2016-06-29 00:00:00
75%	2017-05-14 00:00:00	2017-05-18 00:00:00
Max	2017-12-30 00:00:00	2018-01-05 00:00:00
Std	—	—

Table 2: Descriptive statistics – Numerical columns

Statistic	Postal Code	Sales	Quantity	Discount	Profit
Count	9,993	9,993	9,993	9,993	9,993
Mean	55,191.58	229.85	3.79	0.156	28.66
Min	1,040	0.44	1	0.00	-6,599.98
25%	23,223	17.28	2	0.00	1.73
50%	56,560	54.48	3	0.20	8.67
75%	90,008	209.94	5	0.20	29.36
Max	99,301	22,638.48	14	0.80	8,399.98
Std	32,065.07	623.28	2.23	0.206	234.27